

# Customer Churn Predictor

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Apoorv Gupta – S24CSEU1122

Mahin Sharma - S24CSEU1100

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Supervisor: Dr. Susmita Das



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# Introduction

Customer retention is a key challenge for any subscription-based service, especially in the streaming industry where user preferences change rapidly and competition is high. Retaining an existing subscriber is not only more cost-effective than acquiring a new one—it also reflects the platform's ability to keep users engaged.

In this project, we focus on predicting customer churn specifically using a Netflix-style dataset. Our model examines various factors such as user activity, watch-time habits, subscription details, and interaction patterns. By analyzing these factors, the system estimates the likelihood of a user cancelling their subscription.

The purpose of this work is to help streaming platforms take proactive steps before a user actually leaves. With accurate churn predictions, businesses can personalize recommendations, offer timely incentives, or improve user experience to retain their audience more effectively.

## Proposed Work / Methodology

The methodology consists of five major stages:

### 1. Data Preprocessing:

Missing values are imputed using median or mode strategies. Label encoding is applied to categorical columns.

### 2. Feature Engineering:

Identifying and including Key features like watch hours, subscription type, payment method, and login frequency.

### 3. Model Training:

The XGBoost algorithm is trained to classify customers as churned or retained. It uses gradient boosting to combine multiple weak learners into a strong ensemble model.

### 4. Graphical Interface:

A Tkinter GUI was developed for usability, which allows user to choose b two modes — CSV input and single-row manual entry.

### 5. Recommendation Engine:

The system suggests three personalized retention strategies for users predicted as potential churners.

## Data Structures and Algorithms Used

The project uses :

Pandas DataFrames: For efficient data management

LabelEncoder: For categorical conversion

Tkinter : For interface rendering

The XGBoost algorithm used forms the prediction engine

The GUI displays tabular results using Treeview widgets, and logic-based rules generate targeted recommendations.

## Result Analysis

The system was tested on unseen data, and results indicated high accuracy and consistency in predictions. Users can visualize churn probabilities and recommendations directly in the GUI. For high-risk customers, suggestions such as offering premium upgrades, personalized reactivation campaigns, and loyalty programs were generated. These features make the model not just a prediction tool but also a practical decision-support system.

## Conclusion

This project demonstrates the effective application of machine learning for customer retention management. By integrating XGBoost with a Tkinter-based GUI, it delivers both technical accuracy and usability. The addition of recommendation features bridges the gap between prediction and actionable insights. Future work includes enhancing personalization through deep learning and developing a cloud-based deployment for scalability.